Chapter 8
Integration of the System

8.1 Calculation of Individual Instruments and Total Error for the System

The accuracy specified for an instrument (eg. 1%) is the error or inaccuracy of any measurement performed with that device. This is assuming that the device is operating within its specifications.

Error calculations become more complex when looking at multiple instruments, or systems with more than one component, or even devices that perform calculations on process measurements.

Throughout this section we will use a signal of 0 - 10V to represent 0 - 100%, this will simplify calculations.

8.1.1 Linear Devices in Series

When combining devices in series, the errors are multiplied. In multiplying any inaccuracy, it is a common mistake to simply multiply out the inaccuracies. The total error takes into account the total measurement with the variation.

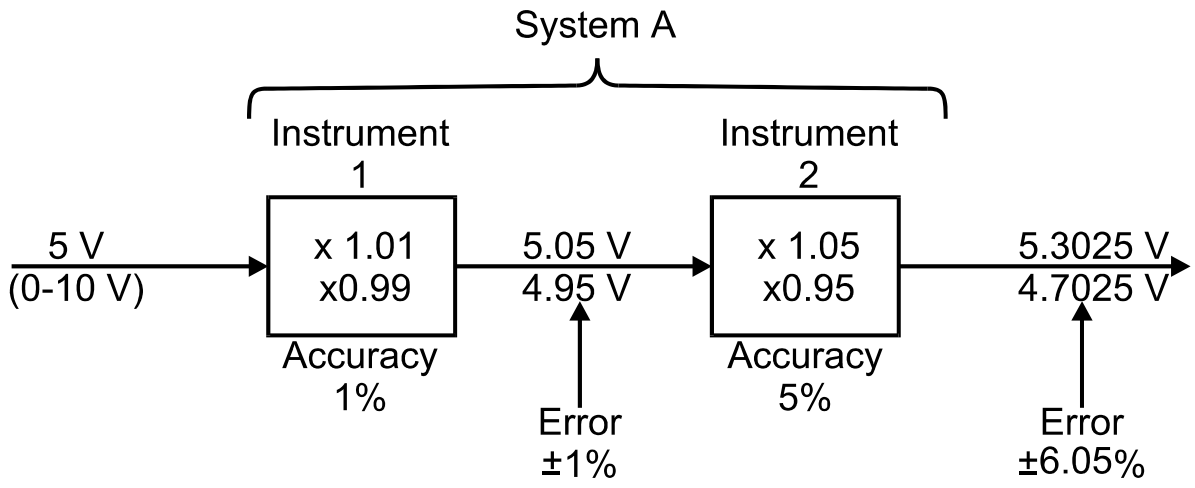


Figure 8.1
Multiple Instruments connected in Series.

If Instrument 1 (as shown) has an incoming signal of 5V, with an accuracy of 1%, then the error in the output will give a signal from 4.95V to 5.05V.

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Note that: 5V x 1.01 = 5.05V maximum measurement due to error
5V x 0.99 = 4.95V minimum measurement due to error

When Instrument 2 receives 5.05V, with its accuracy of 5%, then the error in the output will give an upper value of 5.05V x 1.05 = 5.3025V. The minimum measurement due to error will be 4.95V x 0.95 = 4.7025.

Note that: 5.05V x 1.05 = 5.3025V maximum measurement due to error
4.95V x 0.95 = 4.7025V minimum measurement due to error

So the maximum output value due to the errors is:

5V x (Instrument 1 error) x (Instrument 2 error)

= 5V x ***1.01 x 1.05***

= 5V x ***1.0605***

= 5.3025V

The error in this system, System A, is ***6.05***%.

8.1.2 Non Linear Devices

One of the most common non linear devices used for process measurement is the flowmeter with differential producers. The flow in these devices is calculated from the square root of the measured pressure in the primary element. Any errors in the differential pressure measurement affects the calculated flow.

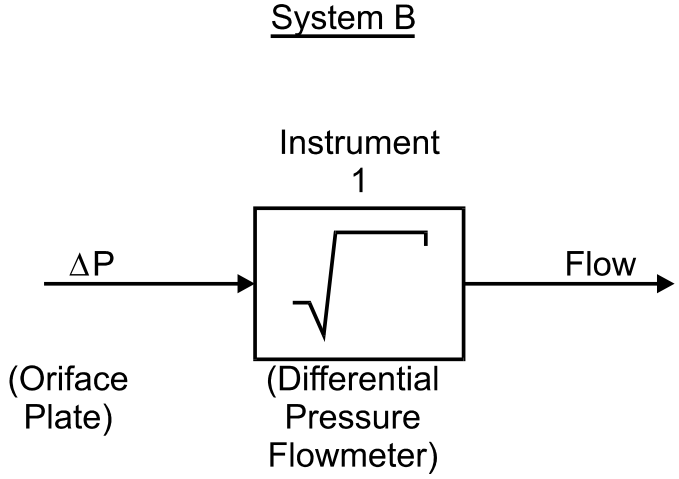


Figure 8.2
Differential Flow Meter.

As the flow is deduced from the square root calculation, the magnitude of the errors are also affected by such a calculation.

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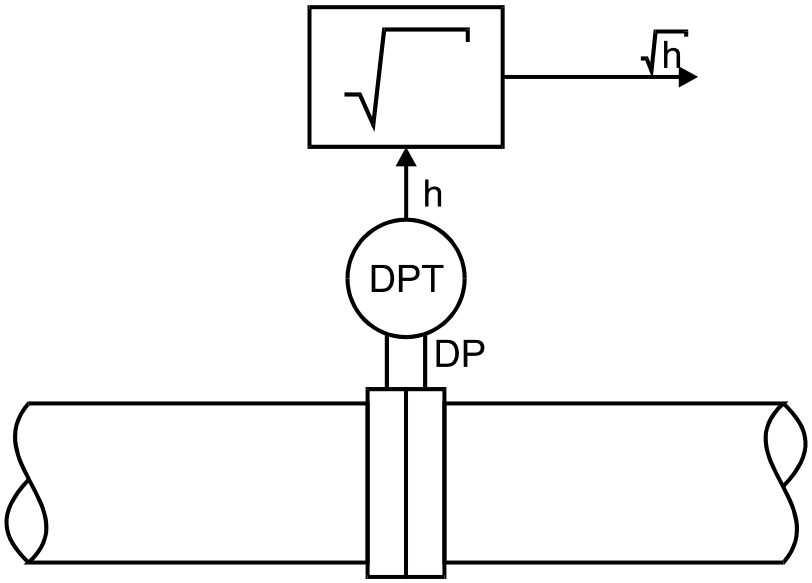


Figure 8.3
Calculation of Total Error.

Actual flow, f (%)	Actual flow, df (0-1.0)	Change in DP, DDP	Change in transmitted signal, dh	Change in calculated flow, dSQRT(h)	Error in calculated flow
100	1.00	2.00	2.02	1.01	1%
80	0.80	1.28	1.30	0.82	2%
60	0.60	0.72	0.76	0.64	4%
40	0.40	0.32	0.41	0.51	11%
20	0.20	0.08	0.26	0.66	46%
10	0.10	0.02	0.25	1.25	115%
5	0.05	0.005	0.26	2.50	245%

Table 8.1
Calculation of error in non-linear device

Column 1: Actual flow, f(%)

This is the actual volumetric flow through the system.

Column 2: Actual flow, df (0-1.0)

This is the normalised flow, simply converted from percent for calculations.

Column 3: Change in DP, dDP

This is the change in DP detected for the specified flow rates.

This is calculated from the flow as follows:

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$DP = f^2$

therefore, the change in DP is represented as,

$dDP = 2f.\ df$

Column 4: Change in transmitted signal, dh

The transmitted signal is the pressure measurement with the combined transmitter error. For this example an error of 0.25% is used.

$dh = dDP$

However, the error needs to be added in. This is done by root mean square as follows,

$dh = SQRT(dDP^2 + e^2)$

Column 5: Change in calculated flow, dSQRT(h)

$dSQRT(h) = dh / 2f$

Column 6: Error in calculated flow

Without an error in the transmitter (e = 0), then column 2 (Actual flow, df) should be equal to column 5 (Change in calculated flow). The differences are due to the error, e=0.25% in the transmitter and can be seen to be quite large at lower flowrates.

8.1.3 Other Non Linear Devices

Other such instruments with non linear calculations are:

- Flumes and rectangular weirs with flows calculated as follows,

$f = h^{3/2}$

- V-notch weirs use a separate formula,

$f = h^{5/2}$

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8.2 Selection Considerations

A comprehensive list for selecting instruments would take into account the following:

- Accuracy
- Reliability
- Purchase price
- Installed cost
- Cost of ownership
- Ease of use
- Process medium, liquid/ stem/ gas
- Degree of smartness
- Repeatability
- Intrusiveness
- Sizes available
- Maintenance
- Sensitivity to vibration

Particular requirements for flow would include:

- Capability of measuring liquid, steam and gas
- Rangeability
- Turndown
- Pressure drop
- Reynolds number
- Upstream and downstream piping requirements

A more systematic approach to selection process measurement equipment would cover the following steps:

Step 1 Application

This is the requirement and purpose of the measurement.

- Monitor
- Control
- Indicate
- Point or continuous
- Alarm

Step 2 Process Material Properties

Many process measuring devices are limited by the process material that they can measure.

- Solids, liquids, gas or steam

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- Conductivity
- Multiphase, liquid/gas ratio
- Viscosity
- Pressure
- Temperature

Step 3 Performance

This relates to the performance required in the application.

- Range of operation
- Accuracy
- Linearity (Accuracy may include linearity effects)
- Repeatability (Accuracy may include repeatability effects)
- Response time

Step 4 Installation

Mounting is one of the main concerns, but the installation does involve the access and other environmental concerns.

- Mounting
- Line size
- Vibration
- Access
- Submergence

Step 5 Economics

The associated costs determine whether the device is within the budget for the application.

- Purchase cost
- Installation cost
- Maintenance cost
- Reliability/ replacement cost

Step 6 Environmental And Safety

This relates to the performance of the equipment to maintain the operational specifications, and also failure and redundancy should be considered.

- Process emissions
- Hazardous waste disposal
- Leak potential
- Trigger system shutdown

Step 7 Measuring Device/Technology

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At this stage the selection criteria is established and weighed up with readily available equipment. A typical example for flow is shown:

- DP	- Orifice plate - Variable area
- Velocity	- Magmeter - Vortex - Turbine - Propeller
- Positive displacement	- Oval gear - Rotary
- Mass flow	- Coriolis - Thermal

Step 8 Supply By Vendors

Limitations may be imposed, particularly with larger companies that have preferred suppliers, in which case the selections may be limited, or the procedure for purchasing new equipment may not warrant the time and effort for the application.

Summary

With any type of measurement (temperature, pressure, level, flow etc.) there is always the possibility of more than one available device being able to satisfy the selection criteria. Conversely there is also the chance that there is no equipment that will perform the task to the required specifications. The order of priority of the selection criteria is very much dependent on the application.

8.3 Testing and Commissioning of the Subsystems

Test procedures vary from one operation to the next, and are generally dependent on how critical the equipment going into service is. In a non-critical application, where the instrument operation will not affect production, the preliminary checks can be performed on the device with it tested insitu.

More critical applications may involve the instrument to be tested on the bench before being placed in operation. Even after installation any outputs should be disabled until the correct operation of the device is proven.

Correct operation requires more than checking that the instrument works. To ensure that it is configured correctly, both for the process measurement and any alarming, trip or indication points should be tested. The interface to the Operator Interface Terminal (OIT) should also be checked.

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Below are steps for a testing procedure for the worst case (most critical) application, with steps easily omitted for less critical cases as required.

- Check correct installation of the instrument
  This also includes grounding and isolation as required. Termination of the wire shields should be at one end only, unless used for an instrument ground path. Verify wire numbering and device tag number.

- Check power supply.
  The reliability of the instrument is also dependent on the supply. Obviously if the main supply fails then that part of the process will not be able to proceed. However, from the instrument point of view, checks should be performed to verify the voltage rating and the proper allocation of breakers or fuses. This will also include the power supplied to I/O cards on such systems.

- Before applying power
  The field wiring should be isolated as well as possible from the digital system until loop commissioning or checkout is complete. This mainly applies to any output devices and can be done by removing fuses, unplugging terminal blocks or lifting isolation links.

- Apply power
  Initial checks should be performed on the system sensibility.
  Check:
  - Indicating lights on system modules
  - Any alarms for validity
  - Temperature inputs for ambient readings
  - Pressure, level and flow for minimum readings
  - System communications
  - Smart instrument communications

- Check loop voltages
  Loop voltages should be checked under load and monitored.

- Check proper calibration of the instrument.
  The vendor usually does this to meet equipment specifications. If required this can be performed remotely with appropriated test equipment before installation into service.

- Loop checkout
  Each loop should be checked for proper operation from the instrument to the digital system. It often depends on the test equipment available and the process as to how much simulation is performed.

- Simulations
  A number of parts of the system can be simulated. This is not always a necessity, but by checking parts of the system faulty components can be eliminated. Such simulations are:
  - The transducer signal, or the input to the instrument

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- The instrument signal, or the input to the controller
- The signal to the control device

Simulations can be performed at a number of values, but the more common being:

- The minimum or lower range value, 0%
- The maximum or upper range value, 100%
- The mid range value, 50%
- Alarming points
- Control points

- Check trips, interlocks and shutdown procedure.
  This is vital before the system is placed into any form of automatic control operation. This is often performed by manually actuating switches and checking that interlocks function as required.

- Automatic Control
  PID controllers should be started in manual mode. Progression to automatic mode should be done under stable conditions and with one loop at a time. The actual procedure for achieving full automatic control of a plant is quite specific to the application. However, by activating automatic control with one loop at a time, it will be much easier to troubleshoot, test and tune the individual loops.

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8.3.1 Prior considerations and budgeting requirements

This test procedure assumes that all associated personnel are familiar with instrumentation and digital systems. Allowances may be required should suitable experience not be available. This may require additional supervision or even the assistance of consulting support.

Loop checkout can be tedious and may require a substantial amount of organisation for large installations. Usually several stages of loop checkout occur before the final check. Sometimes, the application may require a more extensive test, such as running nitrogen, water, or a similar safe process medium through the facility and checking the control operation. The suitability of the medium used for testing must be checked, as it may not respond as the correct sensing material would, or may not work with the instruments engineered for the actual process.

The project budget and schedule will dictate the extent to which the system can be tested, but thorough testing increases familiarity of personnel involved and the chances for a smooth and safe start-up.

It is also good practice to involve the facility operators and maintenance personnel in the later stages of the installation and start-up of the process, this includes loop commissioning. This eases the handover to the facility personnel who will be using the system and are ultimately are responsible for production.

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Tips and Tricks



Tips and Tricks



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